

Impacts of Infrastructure Outages and Cascading Effects on Critical Facility Accessibility: A Case Study of Trier

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Abstract. Critical infrastructures (CIs), such as hospitals, fire stations, and utility networks, are vital for urban functionality yet remain highly vulnerable to extreme environmental hazards. While existing research frameworks typically evaluate either transport-related road accessibility or utility failure cascades in isolation, they fail to capture how cascading outages dynamically degrade spatial emergency service delivery. In this paper, we propose a dynamic framework linking the Robustness of Accessibility (RoA) index with time-dependent modeling of cascading infrastructure failures. We extend the RoA metric across a continuous disaster horizon and introduce a four-state Finite State Machine (FSM), spanning operational, depleting, dead, and rebooting states, to govern facility operability based on power and water grid dependencies, backup buffer capacities, and recovery hysteresis. Interdependent utility assets along subterranean corridors are topologically coupled to critical Points of Interest (POIs) using Network-Constrained Nearest-Neighbor (NCNN) graph routing. We implement a decoupled, two-phase geospatial simulation pipeline and evaluate a 336-hour HQ100 flood scenario in Trier, Germany, across four complexity tiers for hospitals and fire stations. The empirical results confirm that transport-only models substantially overestimate urban disaster resilience, yielding a 6.38% higher cumulative accessibility index than the compound failure model. Power outages emerge as the primary systemic cascading bottleneck, whereas backup buffers grant critical operational bridging during flood onset. Furthermore, facility reboot latencies induce a pronounced 48-hour post-flood recovery lag, and topological coupling exposes severe vulnerabilities in physically unflooded peripheral districts due to upstream utility inundation. Ultimately, our findings provide actionable insights for municipal disaster management and climate-resilient infrastructure planning.

Keywords: Critical Infrastructure · Cascading Failures · Robustness of Accessibility · Finite State Machine · Urban Resilience · Flood Hazard · Network-Constrained Routing.

1 Introduction

Modern societies depend heavily on the continuous functioning of critical infrastructure (CI), including water, healthcare, energy supplies, and transportation networks [7,4]. These systems are characterized by complex functional interdependencies, meaning that disruptions in one sector can propagate to others, potentially causing widespread cascading failures [6,2]. When evaluating crisis scenarios, it is therefore insufficient to consider infrastructures in isolation. Instead, one must understand how failures spread over time and across sectoral boundaries [7].

A critical aspect of infrastructure resilience is the ability to provide essential services to the population. This depends not only on the operational status of a **Point of Interest** (POI), such as a hospital, but also on its accessibility via the road network. Kaub et al. recently introduced the **Robustness of Accessibility** (RoA) approach to quantify how road network disruptions affect the accessibility of individual POIs [5]. However, existing research often treats traffic-related accessibility and the temporal propagation of failures in utility grids separately. While the RoA focuses on road-based resilience, other models, such as those by Lickert et al., focus on functional cascades within interdependent utility infrastructures [7].

Despite these advancements, a significant research gap remains: it is currently unclear how cascading failures in utility systems and traffic-related accessibility jointly impact the overall resilience of a region over time. This leads to the central research question of this work: *How do cascade effects in utility infrastructures influence the time-dependent reachability (RoA) of critical POIs?*

To address this question, we examine the following hypotheses:

- **H1:** The inclusion of power and water outage cascades results in a lower resilience assessment than a purely road-based RoA analysis would suggest.
- **H2:** By modeling *lifetimes*, the period during which a CI remains functional despite the loss of a supply service (e.g., via emergency power), it is possible to identify critical time windows for disaster management.

The primary goal of this paper is to methodologically link the road-based RoA assessment with time-dependent modeling of cascading failures. We extend the RoA index to incorporate the operational capability of POIs based on their electricity and water supply status. Using the *lifetime* concept [7], we classify infrastructures as inoperable once bridging times have elapsed, providing a dynamic resilience analysis.

The remainder of this paper is structured as follows: Section 2 reviews related work and identifies the research gap. Section 3 introduces our methodology, including the functional RoA extension and the Finite State Machine (FSM) model. Section 4 details the technical implementation and the Trier case study. Section 5 evaluates the results and tests our hypotheses. Section 6 discusses findings and limitations, while Section 7 concludes the paper and offers an outlook on future research directions.

2 Fundamentals and Related Work

Existing methods for assessing CI resilience consider various behavioral aspects during disruptions [3,5,7]. Resilience is not uniformly defined: Calvert and Snelder see it as a system’s ability to cope with disruptions and restore functionality [3], while Kaub et al. focus on road *serviceability* [5]. Bruneau et al. highlight its time-dependent nature, where performance decreases during disruptions and gradually recovers [1]. We follow this temporal perspective, evaluating functional losses and subsequent recovery.

2.1 Interdependencies and Cascading Failures

CI are complex, interconnected networks [4]. Interdependencies are reciprocal relationships where one infrastructure’s state influences another [6]. These can be physical, cyber-related, or geographic [4]. Failures causing components of another infrastructure to fail are referred to as cascading failures [6], which Buldyrev et al. show can propagate recursively [2]. Lickert et al. investigate cascades triggered by power outages, noting that infrastructures may maintain functionality via emergency supply (e.g., hospital generators) for a limited duration, termed *lifetimes* [7]. Following these lifetimes and *repair times*, systems transition between *operative* and *dysfunctional* states [7].

2.2 Road Network Resilience and Accessibility

Emergency services depend on road network *serviceability*, the availability of connections determining travel distances and costs [5]. Disruptions reduce this serviceability, increasing travel times [5]. While the *Link Performance Index for Resilience* (LPIR) evaluates segments, it requires detailed data [3]. Similarly, the *Loss of Serviceability* (LoS) index evaluates resilience by comparing the shortest paths to analyze detours [5]. However, LoS fails when nodes become isolated, as the distance becomes infinite and the index loses its explanatory value [5].

Kaub et al. introduced the *Robustness of Accessibility* (RoA) to evaluate entire urban networks [5]. RoA compares the shortest path lengths in an intact network ($c(s_j, d_j)$) to a disrupted version ($c'(s'_j, d'_j)$). For each node pair, the ratio of these path lengths is weighted by $a_{c,j}$ and combined:

$$\text{RoA}(\mathcal{C}) = \sum_{j=1}^m \left(a_{c,j} \cdot \frac{c(s_j, d_j)}{c'(s'_j, d'_j)} \right) \quad (1)$$

RoA values range from 0 (complete failure) to 1 (full resilience) [5]. If a destination becomes completely unreachable, the ratio is set to 0 [5]. While Kaub et al. focus on road accessibility, they omit functional facility states [5]. Conversely, Lickert et al. model functional cascades without considering road accessibility [7]. This paper bridges this gap by extending the RoA index with a time-dependent functional dimension, integrating CI cascading failures with the resulting impacts on urban accessibility.

3 Methodology: Time-Dependent Functional RoA

To quantify the impact of cascading infrastructure failures on urban resilience, we extend the static RoA index into a dynamic framework. This framework incorporates temporal flood progression, road-network coupling, utilities like power and water, and facility resilience managed via a finite state machine (FSM).

3.1 Mathematical Framework of RoA(t)

Time Deterministic RoA: While preserving the standard RoA structure (Equation 1), downstream accessibility is dynamically modified by facility functionality, directly governing destination reachability. We operationalize this as a time-dependent index, $\text{RoA}(\mathcal{C}, t)$, evaluated hourly over the simulation horizon T . Consequently, system states are parameterized continuously over time rather than denoted by static prime notation (c vs. c'), where t_0 is the baseline and t is any step in the simulation. The new RoA formula looks as follows:

$$\text{RoA}(\mathcal{C}, t) = \sum_{j=1}^m \left(a_{\mathcal{C},j} \cdot \frac{c(s_j, d_j, t_0)}{c(s_j, d_j, t)} \right) \quad (2)$$

$\text{RoA}(\mathcal{C}, t)$ Functional area-wide accessibility index at time t in the simulation.
 $\mathcal{C} / m$ Set ($\mathcal{C}$) and count (m) of origin-destination POI connections.
 $a_{\mathcal{C},j}$ Weighting factor of routes j for POI $\mathcal{C}$ subject to $\sum_{j=1}^m a_{\mathcal{C},j} = 1$.
 $c(s_j, d_j, t_0)$ Travel cost from origin s_j to destination d_j at initial time t_0 .
 $c(s_j, d_j, t)$ Travel cost from origin s_j to destination d_j at simulation time t .

Time-Integrated RoA over T : Because $\text{RoA}(\mathcal{C}, t)$ is evaluated continuously over the course of the simulation rather than as a static snapshot, total resilience over the crisis period is captured by the time-integrated RoA (RoA_{Int}):

$$\text{RoA}_{\text{Int}}(\mathcal{C}) = \frac{1}{t_N - t_0} \int_{t_0}^{t_N} \text{RoA}(\mathcal{C}, t) dt \quad (3)$$

$\text{RoA}_{\text{Int}}(\mathcal{C})$ Cumulative time-integrated resilience index over the disaster horizon.
 $\mathcal{C}$ Set of evaluated origin-destination POI connections.
 $[t_0, t_N]$ Analysis period from the start (t_0) to the end (t_N) of the simulation.
 $\frac{1}{t_N - t_0}$ Normalizing the area under the resilience curve to $[0, 1]$.
 $\text{RoA}(\mathcal{C}, t)$ Time-dependent accessibility index of connections $\mathcal{C}$ at time t .
 dt Differential element for integration over time t .

3.2 POI Health: Finite State Machine and State Transitions

While physical road availability is a necessary condition for accessibility, it is insufficient to guarantee operational functionality. Facilities such as hospitals and fire stations depend heavily on external utilities (power and water). When utility grid connections fail, internal backup systems allow temporary operation. These states and transitions need to be governed by a Finite State Machine (FSM), which also bridges the gap between Kaub et al. and Lickert et al.

FSM States: To capture this dynamic functionality, we model each POI i using a four-state FSM where at any time step t , the state of a facility is defined as $\sigma_i(t)$, which determines an operational binary indicator $f_i(t) \in \{0, 1\}$ indicating whether a facility can actively accept traffic and deliver service:

$$\sigma_i(t) \in \left\{ \begin{array}{l} \text{Oper, Depl.,} \\ \text{Dead, Rebo.} \end{array} \right\} \quad (4) \quad f_i(t) = \left\{ \begin{array}{ll} 1 & \text{if } \sigma_i(t) \in \{\text{Oper., Depl.}\} \\ 0 & \text{if } \sigma_i(t) \in \{\text{Dead, Rebo.}\} \end{array} \right. \quad (5)$$

Operational All external utility connections are active and reserve buffers recharge. The facility is fully functional ($f_i(t) = 1$).

Depleting At least one utility connection has failed while internal reserves bridge the outage. The facility remains functional ($f_i(t) = 1$).

Dead At least one reserve buffer is fully exhausted or the facility site is directly flooded. The facility is non-functional ($f_i(t) = 0$).

Rebooting All utility supplies are restored or buffer reserves are above thresholds. The facility is non-functional ($f_i(t) = 0$).

FSM State Transitions and Hysteresis: FSM state transitions are governed by resource availability, continuous flood exposure, and hysteresis rules, as summarized in Table 1. Here, $\mathbf{p}_i$ denotes the geographic coordinates of facility i , with $\mathbf{p}_i \in \text{FloodPolygon}(t)$ indicating direct site inundation at time t . Resource dependencies, including utility status $u_{i,r}(t)$, reserve buffers $b_{i,r}(t)$, and maximum capacities B_r for $r \in \{\text{power, water}\}$, are detailed in the subsequent *FSM Buffer Dynamics* section. The parameter $\theta = 0.15$ enforces hysteresis by defining the minimum relative buffer capacity required to initiate rebooting without full grid restoration, preventing rapid failure-recovery cycling (chattering).

Table 1: State transitions and guard conditions governing facility FSM updates.

Current State	Next State	Transition Guard Condition
Operational	Dead	$\mathbf{p}_i \in \text{FloodPolygon}(t)$
	Depleting	$\exists r \in \{\text{power, water}\} : u_{i,r}(t) = 0$
Depleting	Dead	$\mathbf{p}_i \in \text{FloodPolygon}(t)$
	Dead	$\exists r \in \{\text{power, water}\} : u_{i,r}(t) = 0 \wedge b_{i,r}(t) = 0$
	Operational	$\forall r \in \{\text{power, water}\} : u_{i,r}(t) = 1$
Dead	Rebooting	$\mathbf{p}_i \notin \text{FloodPolygon}(t) \wedge (\forall r \in \{\text{power, water}\} : u_{i,r}(t) = 1 \vee b_{i,r}(t) > \theta \cdot B_r)$
Rebooting	Dead	$\mathbf{p}_i \in \text{FloodPolygon}(t)$
	Dead	$\exists r \in \{\text{power, water}\} : u_{i,r}(t) = 0 \wedge b_{i,r}(t) \leq \theta \cdot B_r$
	Operational	$t_{\text{reboot}} \geq T_{\text{repair}} \wedge (\forall r \in \{\text{power, water}\} : u_{i,r}(t) = 1)$
	Depleting	$t_{\text{reboot}} \geq T_{\text{repair}}$

FSM Buffer Dynamics: Each POI i maintains independent reserve buffers $b_{i,r}(t)$ for each resource $r \in \{\text{power, water}\}$ with maximum capacity B_r . During grid outages ($u_{i,r}(t) = 0$), reserves deplete at the rate λ_r . When grid connectivity is restored ($u_{i,r}(t) = 1$), reserves recharge at the rate μ_r , while $f_i(t) = 1$:

$$b_{i,r}(t+1) = \begin{cases} \max(0, b_{i,r}(t) - \lambda_r) & \text{if } u_{i,r}(t) = 0 \wedge f_i(t) = 1 \\ \min(B_r, b_{i,r}(t) + \mu_r) & \text{if } u_{i,r}(t) = 1 \wedge f_i(t) = 1 \end{cases} \quad (6)$$

$b_{i,r}(t)$, B_r Reserve level and max capacity for resource r and POI i at time t .

λ_r , μ_r Depletion and recharge rates per unit time for resource r .

$u_{i,r}(t)$ Binary grid connection status for POI i and resource r at time t .

$\max, \min$ Clamping bounds enforcing physical reserve limits $[0, B_r]$.

Integrated FSM Workflow: The overall interaction between utility connectivity, reserve buffer dynamics, direct flood overrides, and reboot hysteresis is synthesized in the finite state machine illustrated in Figure 1.

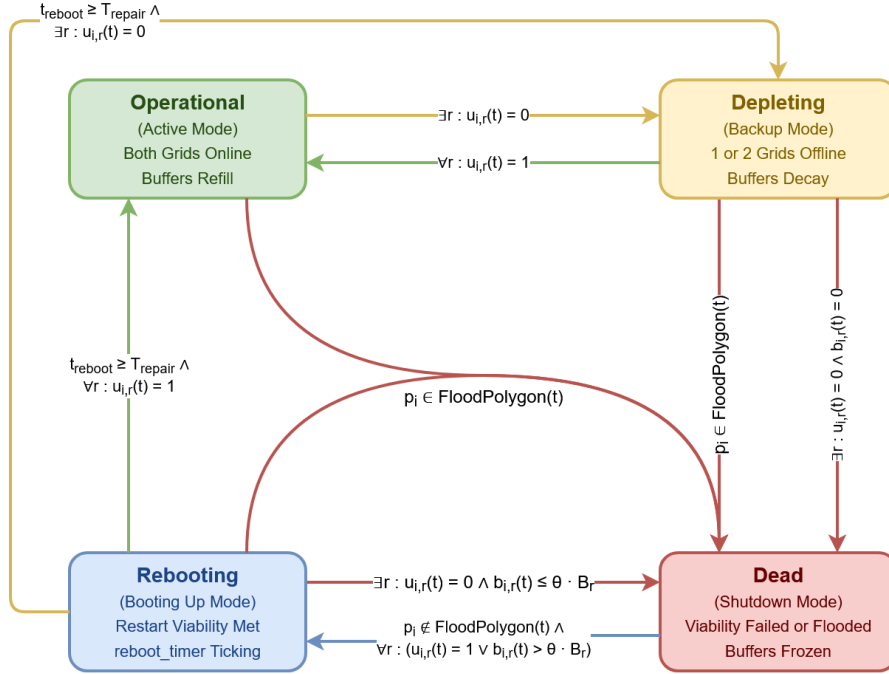


Fig 1: Finite state machine governing POI functional status based on utility connectivity, reserve dynamics, hysteresis thresholds, and environmental flood overrides.

3.3 Infrastructure Coupling via NCNN

Critical utility networks (power and water) physically follow street corridors underground, despite the individual supply assets and POIs often being located off the road network itself, as highlighted by Okabe et al. [8]. To capture topological barriers without assuming unconstrained Euclidean connectivity, utility dependencies are established using Network-Constrained Nearest-Neighbor (NCNN) mapping over an edge-weighted undirected street graph $G = (V, E)$. This framework accurately models underlying conduit pathways, enables spatial visualization within the simulation engine, and bridges continuous geographic space with discrete graph topology by projecting off-network entities onto their nearest street network components. Each POI i is mapped to its topologically closest supply asset $s^*(i, r)$ for resource type $r \in \{\text{power, water}\}$ via a multi-source shortest-path search:

$$s^*(i, r) = \arg \min_{s \in \mathcal{I}_r} \text{dist}_G(\pi(\mathbf{p}_i), \pi(s)) \quad (7)$$

- $s^*(i, r)$ Assigned asset of resource type r topologically closest to facility i .
- $\arg \min_{s \in \mathcal{I}_r}$ Minimization operator selecting the asset s yielding shortest path.
- $\mathcal{I}_r$ Set of active, operational supply assets for resource r .
- $\text{dist}_G(u, v)$ Shortest network path distance between snapped start vertex u and end vertex v on G .
- $\pi(\mathbf{p}_i), \pi(s)$ Snapping operators projecting off-network POI ($\mathbf{p}_i$) and asset (s) coordinates to graph vertices $u, v \in V$

This topological assignment is static and happens only once at the start of the simulation. A dynamic assignment could be implemented by governing the dynamic grid status $u_{i,r}(t)$ within the FSM:

$$u_{i,r}(t) = \begin{cases} 0 & \text{if } \mathbf{p}_{s^*(i,r)} \in \text{FloodPolygon}(t) \\ 1 & \text{otherwise} \end{cases} \quad (8)$$

If the assigned supply asset $s^*(i, r)$ is inundated, the utility connection is severed ($u_{i,r}(t) = 0$), initiating buffer depletion ($b_{i,r}$) in the POI's state machine.

4 Implementation

Evaluating dynamic accessibility across thousands of origins over 336 hourly timesteps is computationally intractable if spatial overlays are computed dynamically at runtime. Our Python 3.11+ pipeline decouples heavy geospatial precomputation from discrete-time simulation for municipal Trier, Germany (117 km²), integrating open geospatial data with official hydrodynamic flood models.

4.1 Hydrodynamic Stage Ingestion and Spatial Preprocessing

Multi-Stage Hydrodynamic Ingestion (HQ_raw): Our hazard model digests empirical hydrodynamic flood stages from the Rhineland-Palatinate Water Management Authority. The dataset (HQ_raw) comprises 29 discrete gauge GeoJSON feature collections (Pegel_09_00m.geojson to Pegel_11_80m.geojson), spanning gauge heights from 9.00 m to 11.80 m (HQ100 peak) in 0.10 m increments [9]. In `flood_interpolation.py`, the ingestion routine projects geometries to WGS84 (EPSG:4326) and UTM Zone 32N (EPSG:25832), dissolving adjacent tiles via `shapely.ops.unary_union` into unified stage polygons $\mathcal{P}_k$ ($k \in [0, 28]$) cached under `data/processed/flood_interpolation/`.

Multimodal Infrastructure and Network Ingestion Infrastructure within Trier ($\sim 117 \text{ km}^2$) is extracted from OpenStreetMap via `OSMnx`:

1. **Road Network (G):** Filtered to drivable roads, yielding an undirected graph of $|V| = 5,817$ nodes with edge traversal costs $c(e) = \text{length}(e)$ measured as physical network distance in meters.
2. **Critical POIs and Utilities:** Destinations comprise 3 acute hospitals and 12 fire stations. Coupled utilities encompass 131 power assets ($\mathcal{I}_{\text{power}}$, substations and plants) and 11 water assets ($\mathcal{I}_{\text{water}}$, waterworks, towers, pumping stations).

Facility footprints are converted to interior representative points $\mathbf{p}_i$ and snapped to nearest network vertices $\pi(\mathbf{p}_i) \in V$ via R-tree queries.

4.2 System Architecture and Execution Pipeline

Modular Architecture and Technology Stack: The pipeline couples five sequential modules: spatial ingestion, hazard overlay analysis, utility dependency mapping, temporal simulation, and scenario evaluation. The software foundation combines `GeoPandas` and `Shapely` (geometry and spatial indexing), `OSMnx` and `NetworkX` (graph extraction and routing), and `NumPy` (vectorized state matrices). As shown in Figure 2, data flows unidirectionally through persistent disk caches into `dynamic_roa.py`.

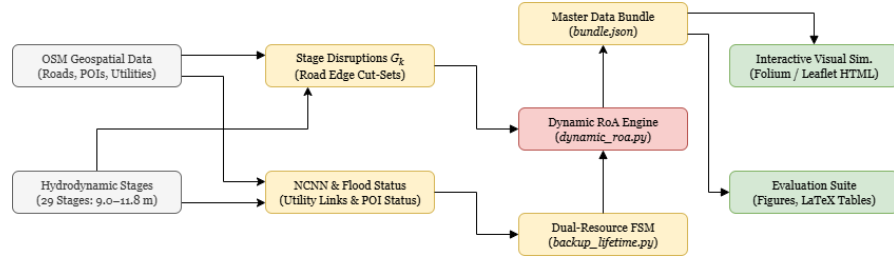


Fig. 2: Modular architecture: Precomputed stage subgraphs and FSM trajectories feed the dynamic RoA simulation engine, producing a master data bundle.

Two-Phase Pipeline Paradigm: Naively intersecting road edges with flood polygons across 1,344 hourly tier evaluations is expensive. Our architecture enforces a two-phase paradigm: **Phase 1 (Spatial Preprocessing)** executes heavy geometric operations, network extraction, boundary aggregation, edge disruption slicing, and utility routing (NCNN), once per study area, caching all cut-sets under `data/processed/`. **Phase 2 (Dynamic Simulation)** loads cached lookups into memory, evaluating FSM transitions and accessibility over 336 hours in seconds without geometric operations.

4.3 Direct Flood Determination and Network Disruption Mapping

Facility Inundation via Interior Representative Points: Determining facility inundation requires robust geometric testing. Standard centroids and bounding envelopes fail on complex non-convex footprints (e.g., L-shaped hospitals): centroids drift into dry courtyards, while bounding boxes trigger false positives when floodwaters overlap parking lots. In `flood_status.py`, the `representative_point()` method guarantees test coordinates lie strictly within the interior: $\mathbf{p}_i \in \text{Int}(\text{Geom}_i)$. Vulnerability across all 29 stages is evaluated via GeoPandas spatial R-trees (`sindex`) in $\mathcal{O}(\log |\mathcal{D}|)$ time and cached as direct flood matrices.

Road Network Edge Disruption and Cut-Sets: Dynamic accessibility degradation depends on road passability. In `stage_disruptions.py`, graph edges are intersected against each stage polygon $\mathcal{P}_k$. To preserve elevated river crossings, edges with a non-null `bridge` tag remain open unless flood heights submerge bridge decks. Flooded edge tuples (u, v, key) form stage cut-sets $E_{\text{flood},k} \subset E$, cached in JSON format. Passable subgraphs $G_k = (V, E \setminus E_{\text{flood},k})$ are instantiated in memory in $\mathcal{O}(|E_{\text{flood},k}|)$ time, bypassing geometric clipping entirely.

Network-Constrained Utility Routing (NCNN): Because utility conduits follow street corridors underground, Euclidean distance distorts real infrastructure dependencies across river barriers. In `ncnn.py`, dependency assignment $s^*(i, r)$ for facility $i \in \mathcal{D}$ and resource $r \in \{\text{power}, \text{water}\}$ is computed on base graph G_0 in UTM Zone 32N (EPSG:25832). To avoid pairwise queries, a multi-source Dijkstra traversal (`nx.multi_source_dijkstra`) runs from all utility assets $\mathcal{I}_r$ simultaneously, mapping each facility node $\pi(\mathbf{p}_i)$ to its nearest utility in $\mathcal{O}(|E| + |V| \log |V|)$ time. Reconstructed paths are cached under `data/processed/ncnn/`, coupling facility FSM states to upstream utility flooding.

4.4 Two-Tier Caching Strategy and Storage Hierarchy

On-Disk Directory Hierarchy and Naming Conventions Intermediate spatial artifacts are persisted under `data/processed/` (Table 2) using deterministic keys `<type>_<place>_<K>_<T>h.<ext>` ($K = 29, T = 336 \text{ h}$), preventing stale cache reads across experiment setups.

Table 2: Persistent on-disk cache hierarchy under `data/processed/`.

Pipeline Phase	Subdirectories	Key Artifacts and Stored Data
Hazard Preprocessing	<code>flood_interpolation/</code> , <code>network/</code>	29 dissolved flood polygons $\mathcal{P}_k$; road cut-sets $E_{\text{flood},k}$.
Spatial Coupling	<code>services/</code> , <code>flood_status/</code> , <code>ncnn/</code>	Snapped POIs; stage inundation matrices; NCNN paths.
Temporal Simulation	<code>backup_lifetime/</code> , <code>roa/</code>	Hourly FSM buffer logs; multi-tier evaluation bundle.

Computational Complexity Reduction: Un-cached accessibility across four tiers over $T = 336$ h (1,344 timesteps) tests every edge against flood polygon $\mathcal{P}(t)$ in $\mathcal{O}(|\mathcal{M}| \cdot T \cdot |E| \cdot |\text{Vertices}|)$ time (> 3 h). Precomputing cuts per stage ($K = 29$) reduces spatial overlays by 97.8%, enabling $\mathcal{O}(1)$ lookups in Phase 2 to evaluate scenarios in < 12 s.

Cache Lifecycle and Invalidation Caching routines expose an execution parameter (`force_recompute`) to recompute directly. Central script `clear_cache.py` flushes `data/processed/` and bytecode caches while preserving raw gauge files in `HQ_raw`.

4.5 Temporal Simulation Engine and Multi-Tier Orchestration

Discrete-Time 336-Hour Simulation Loop In `dynamic_roa.py`, time advances hourly ($t \in [0, 335]$) across five crisis phases, mapping progress $p(t)$ to gauge stage $k(t)$. Facility state $\sigma_i(t)$ in `backup_lifetime.py` spans four modes (*Operational*, *Depleting*, *Dead*, *Rebooting*): (1) interior flooding overrides buffers to *Dead*; (2) severed utilities trigger *Depleting*, drawing power (b_{power}) or water (b_{water}) at rates λ_r ; and (3) restoration requires hysteresis recharge $b_{i,r} > 0.15B_r$ and reboot latency T_{repair} (24 h fire stations, 48 h hospitals) before active delivery ($f_i(t) = 1$).

Multi-Tier Scenario Execution and Payload Packaging Demand is sampled across $m = 3,972$ origin nodes (50 m grid). The multi-tier runner evaluates four resilience tiers under uniform recovery parameters ($\theta = 0.15$ hysteresis threshold, reboot delay $T_{\text{repair}} = 24$ h for fire stations and 48 h for hospitals):

- **Level A:** Infinite power and water buffers ($B_{\text{power}} = B_{\text{water}} = \infty$) across all facilities, isolating pure road network disruptions as a baseline.
- **Level B1:** Infinite water buffers ($B_{\text{water}} = \infty$) with finite power reserves ($B_{\text{power}} = 24$ h for fire stations, 72 h for hospitals).
- **Level B2:** Infinite power buffers ($B_{\text{power}} = \infty$) with finite water reserves ($B_{\text{water}} = 12$ h for fire stations, 48 h for hospitals).
- **Level C:** Compound power and water buffers combining the finite reserve constraints of Levels B1 and B2.

Multi-source Dijkstra from active facilities $\mathcal{D}_{\text{active}}(t)$ on $G_{k(t)}$ runs in $\mathcal{O}(|E| + |V| \log |V|)$ time, memoized under $(k(t), \mathcal{D}_{\text{active}}(t))$. Outputs are serialized into `multi_tier_roa_bundle_Trier_Germany_336h.json`, storing RoA(t) curves, FSM logs, and district metrics for downstream evaluation.

5 Results and Evaluation

The following section presents the empirical findings of the multi-tier simulation across the 336-hour flood horizon. We evaluate how infrastructure coupling impacts integrated accessibility robustness (RoA_{Int}) and test hypotheses on resilience overestimation, buffer dynamics, and recovery lag.

5.1 Scope and Methodological Baseline

Our Level A (Road-Only) baseline yields $RoA_{\text{Int}} = 84.58\%$ and a systemic nadir of $RoA_{\text{Sys,min}} = 38.41\%$ during peak HQ100 inundation, diverging from the 70.1 % regional benchmark by Kaub et al. [5]. While their study spans 26,853 nodes and 44 fire stations across surrounding rural counties, our model focuses on municipal Trier (117 km², 5,817 nodes, 12 fire stations). This focused scope eliminates peripheral highway bypasses and suburban redundancy, exposing core urban vulnerabilities. Combined with localized hydrodynamics, Level A establishes a realistic transport baseline to isolate compound utility coupling (Level C).

5.2 Multi-Tier Quantitative Results and Simulation Timeline

Table 3 summarizes aggregate metrics across all four complexity tiers, and Figure 3 illustrates the systemic accessibility timeline across five disaster phases:

1. **Pre-Event** ($0 \leq t < 48 \text{ h}$): Undisrupted baseline conditions.
2. **Rising Limb** ($48 \leq t < 144 \text{ h}$): Floodwaters rise (9.00 m $\rightarrow$ 11.80 m), closing roads and submerging substations while buffers sustain operations.
3. **HQ100 Peak Plateau** ($144 \leq t < 192 \text{ h}$): Buffer exhaustion forces facilities offline, reaching nadir (38.41 % Level A vs. 37.29 % Level C).
4. **Recession Limb** ($192 \leq t < 288 \text{ h}$): Floodwaters recede, Level A recovers synchronously with road clearance, while Level C remains incapacitated.
5. **Post-Event Recovery** ($288 \leq t \leq 336 \text{ h}$): Level A fully restores at hour 288, while Level C displays a 48-h recovery lag from reboot delays.

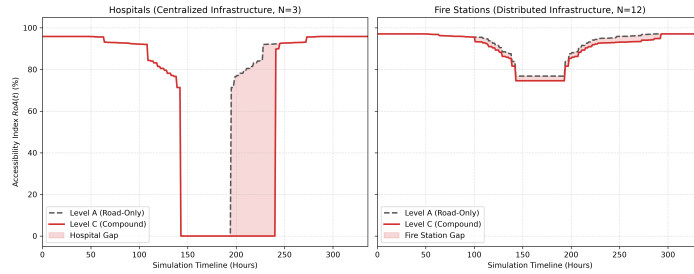


Fig. 3: Temporal evolution of RoA for Level A and Level C over the 336-h horizon. Shading indicates the cumulative functional overestimation gap.

Table 3: Quantitative summary of multi-tier simulation results for Trier.

Level	RoA_{Int}	$RoA_{\text{Sys,min}}$	Hosp. Nadir	Fire Nadir
Level A	84.58%	38.41%	0.00%	76.82%
Level B1	78.20%	37.29%	0.00%	74.59%
Level B2	78.84%	38.41%	0.00%	76.82%
Level C	78.20%	37.29%	0.00%	74.59%

Intermediate tiers reveal power grid failure (Level B1) as Trier’s primary cascading bottleneck: Level B1 matches Level C identically ($RoA_{\text{Int}} = 78.20\%$, nadir 37.29%), whereas Level B2 maintains higher resilience ($RoA_{\text{Int}} = 78.84\%$, nadir 38.41%). This indicates that water networks are less vulnerable to flooding than power infrastructure. This is attributed to the location of water towers, which are situated farther away from the Mosel than power infrastructure and are therefore less likely to be affected by flooding.

5.3 Hypothesis Testing and Validation

Functional Gap and Resilience Overestimation: Hypothesis H1 posits that transport-only models overestimate disaster resilience by ignoring cascading infrastructure dependencies. This is confirmed: Level C reveals a 1.12% drop in peak nadir (37.29% vs. 38.41%) and a 6.38% reduction in cumulative resilience ($RoA_{\text{Int}} = 78.20\%$ vs. 84.58%), demonstrating significant resilience overestimation. The main contributors to this difference are the fire stations, where additional facilities are lost due to power outages, whereas hospitals lose all functionality across all levels due to inundation, as shown in Figures 3 and 5.

Temporal Buffers and Post-Flood Recovery Hysteresis: Hypothesis H2 posits that facility buffers grant operational delay during onset, while restart requirements produce recovery hysteresis. This is confirmed: emergency power buffers delay facility collapses by up to 24 hours during flood rise. This 48-hour recovery lag validates the operational penalty of FSM reboot latencies ($T_{\text{repair}} \in \{24 \text{ h for fire stations}, 48 \text{ h for hospitals}\}$). This buffer delay can be observed in Figure 3 while the recovery lag can be observed in Figure 3.

5.4 Spatial Vulnerability: Municipal Patterns and Service Disparity

Figure 4 maps accessibility loss (ΔRoA_j) across Trier at HQ100, comparing the transport baseline (Level A) to compound failure (Level C), where river inundation severs primary transit arteries. Because healthcare facilities ($N = 3$) cluster along the river axis, road closures eliminate access entirely, reducing hospital accessibility to 0.00% across all tiers, access is thus strictly transport-dominated. Conversely, fire stations ($N = 12$) distributed across municipal districts provide redundancy, preserving a 76.82% nadir in Level A. Under Level C, utility cascades reduce this to 74.59% (Figure 5), though Ruwer-Eitelsbach remains operational throughout the buffer despite power-station flooding.

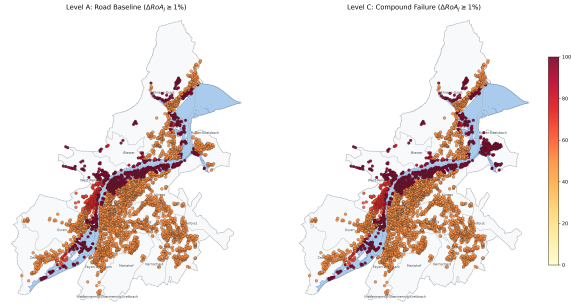


Fig. 4: Spatial accessibility loss (ΔRoA_j) across Trier at peak flood (HQ100): Level A road baseline (left) vs. Level C compound failure (right).

5.5 District Vulnerability: The Ruwer-Eitelsbach Cascade

District analysis reveals that divergence between Level A and Level C concentrates in Ruwer-Eitelsbach. In all 18 other districts, the peak gap is negligible as facilities stay connected or are already road-isolated. Ruwer-Eitelsbach exhibits a 31.99 % mean gap across 140 nodes (peaking at 50.00 %, Figure 5) via a four-step cascade: (1) **Topographical Immunity**: The local fire station stays dry above HQ100 (Level A assumes 100 % operational). (2) **Off-Site Inundation**: In Level C, its upstream valley substation submerges early during flood rise. (3) **Buffer Exhaustion**: After 24 hours, emergency generator fuel depletes, transitioning the station to Dead. (4) **Compound Cutoff**: Concurrently, arterial roads to central Trier submerge, leaving the district isolated.

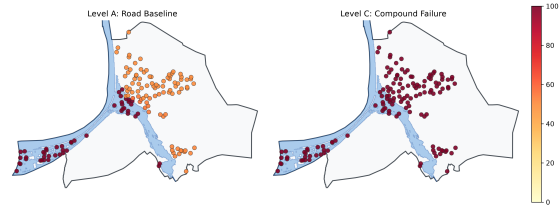


Fig. 5: Heatmap of accessibility loss in Ruwer-Eitelsbach at peak flood: Level A baseline (left) vs. Level C compound failure (right).

5.6 Synthesis and Operational Implications

These empirical findings demonstrate that single-sector resilience models severely underestimate disaster risk. By capturing infrastructure coupling, our framework exposes two critical blind spots: localized utility cascades in physically dry districts and a 48-hour restoration lag. Emergency planners must therefore prioritize flood-hardened power feeds and sustained recovery refueling logistics.

6 Discussion

The extension of the RoA index through infrastructure coupling and state-based modeling provides a more granular understanding of urban vulnerability than road-only assessments.

6.1 The Impact of Utility Interdependencies

A key finding from the multi-tier evaluation is the spatial decoupling of flood hazard and accessibility loss. In Level A, accessibility loss is localized to the immediate inundation area. However, in Levels B1 and C, POIs located on high ground can become functionally dead if their snapped utility infrastructure, often located in lower-lying, flood-prone street corridors, is compromised. This highlights the importance of topological coupling: traditional Euclidean mapping would likely underestimate these cascading risks by ignoring the spatial constraints of utility networks which typically follow street corridors.

6.2 Resilience Buffers and Hysteresis

The FSM logic demonstrates that facility resilience is a function of both buffer capacity and restoration speed. The inclusion of the 15% hysteresis threshold and the reboot delay (T_{repair}) introduces a significant temporal lag in the recovery phase. Even after floodwaters recede and utility connections are restored, the $\text{RoA}(t)$ curve remains depressed while facilities recharge and reboot. This suggests that urban resilience planning must focus not only on physical flood protection but also on increasing internal buffer capacities and streamlining facility reboot protocols.

6.3 Model Limitations

While the proposed framework improves upon static models, several limitations remain:

- **Binary Utility States:** Power and water are modeled as either fully connected or fully lost, ignoring partial failures or pressure drops.
- **Static Infrastructure Mapping:** The coupling between POIs and infrastructure is assumed to be fixed. In reality, utility networks may have redundant paths or the ability to reroute power.
- **Origin-Destination Simplification:** Citizen samples are modeled as points without considering the dynamic behavior of the population during an evacuation.

7 Conclusion

This study extended the RoA index to capture interdependencies between urban transportation and utility infrastructure during extreme floods. Integrating a four-state finite state machine with road-network-constrained coupling advances evaluation beyond simple inundation models to full functional resilience. Findings from the Trier case study demonstrate that:

1. **Cascading failures** significantly amplify accessibility loss, frequently impacting areas far beyond the immediate flood zone.
2. **Internal facility buffers and reboot delays** create critical temporal lags in service restoration.
3. **A multi-tier evaluation approach** is essential to isolate specific network vulnerabilities and compound risks.

Future work should incorporate dynamic utility routing, probabilistic failure modeling, and real-time population movement data. Ultimately, this framework equips planners and emergency managers to quantify service resilience and prioritize targeted infrastructure hardening against climate hazards.

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Declaration of Authorship and AI Usage

Forschungspraktikum Wirtschaftsinformatik – Sommersemester 2026

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We hereby declare that we have written this thesis independently and have not used any sources or aids other than those specified. All direct or indirect content and ideas from external sources are acknowledged as references. We confirm that we have not previously submitted this thesis in part or whole for any other academic credit nor has the work been submitted to any other examination office and has not been published.

We confirm that we have only used text or other content-generating aids based on generative artificial intelligence (GenAI) in the manner permitted in writing by the examiner and that we have explicitly marked GenAI-generated passages.

When using GenAI tools, we have indicated these with their product name and listed them in full in an appendix.

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Thesis title: Impacts of Infrastructure Outages and Cascading Effects on Critical Facility Accessibility: A Case Study of Trier

Programme: Forschungspraktikum - Wirtschaftsinformatik (SoSe 2026)

Examiner: Berndt, Jan Ole, Dr.-Ing. and Timm, Ingo J., Univ.-Prof. Dr.

This is an achievement for a:

- ☐ Bachelor's thesis
- ☐ Master's thesis
- ☒ other examination achievement for module (term paper etc.)

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Place, Date


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Overview of Used Generative AI Tools and Digital Aids
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In accordance with the guidelines of the *Forschungspraktikum* and the official regulations of Trier University, we hereby declare the comprehensive usage of generative artificial intelligence (GenAI) models and digital tools during this research project.

1. Utilized GenAI Models and Platforms:

- **JetBrains Junie** (integrating Google Gemini 3.6 Flash, 3.7 Flash, and 3.8 Flash)
- **OpenAI ChatGPT** (utilizing ChatGPT-5.6 Sol and ChatGPT-6 Astra)

2. Literature Comprehension and Background Research: Literature research was conducted manually using literature databases and a structured snowballing approach without automated AI search agents. GenAI tools were utilized strictly to aid in explaining and comprehending complex domain concepts, mathematical formulations (e.g., RoA calculations), and underlying methodology from source literature. All conceptual insights gained were independently verified against the original publications.

3. Critical Reflection on Research Questions and Methodology: Following our independent initial formulation of the research questions and experimental methodology, GenAI models were employed to critically review and stress-test our design. Specifically, the tools were prompted to evaluate the precision, scope, and time feasibility of our research questions regarding infrastructure cascading failures. Suggestions were critically evaluated, and final adjustments were made autonomously.

4. Software Development and Code Implementation: During software development, GenAI served extensively as an interactive implementation assistant embedded within the IDE. We independently designed the overall software architecture, FSM logic, network routing algorithms, data models, and pipeline execution flow. Prior to coding, we formulated explicit execution plans, functional specifications, expected outputs, and testing strategies for each component. GenAI was then instructed with these highly specific technical directives to write discrete functions and classes, draft geospatial data preprocessing scripts, build visualization components, translate our predefined concepts into code, and assist in navigating legacy codebases. Every line of AI-assisted code was manually reviewed, understood, refactored, and tested against our predefined validation criteria to ensure complete execution correctness.

5. Academic Writing, Style Refinement, and Proofreading: The entire manuscript was initially written and drafted independently by the authors. GenAI was subsequently used exclusively as a post-writing editing aid to correct grammar, refine vocabulary, improve sentence flow, and polish the academic tone **throughout the entirety of the paper**. Text suggestions were critically evaluated on an individual, sentence-by-sentence basis before integration. No scientific content, statistical interpretations, or manuscript sections were autonomously generated by AI.

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